Physical Grounding of the Critical Coherence Index:

Entropy Production and Structural Information in Nonlinear Field Dynamics

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Abstract

The Critical Coherence Index (CCI), introduced in the companion paper [4] as a dimensionless instability ratio for nonlinear field systems, is here given a physical grounding in terms of coarse-grained histogram activity and structural information.

We propose that the CCI approximates the ratio between the coarse-grained value-distribution turnover rate $\dot{S}_{\mathrm{cg}}$ and the nearest-neighbour structural mutual information I_{nn}:

$$\mathcal{C}_{\mathrm{CCI}} \approx \frac{\dot{S}_{\mathrm{cg}}}{I_{\mathrm{nn}} + \varepsilon}.$$

To test this hypothesis, we measure $\dot{S}_{\mathrm{cg}}$, I_{nn}, and $\mathcal{C}_{\mathrm{CCI}}$ across an ensemble of four configuration classes in a one-dimensional ϕ^4 scalar field theory: vacuum states, kink solitons, localised wave packets, and chaotic configurations.

The Spearman rank correlation between $\mathcal{C}_{\mathrm{CCI}}$ and the ratio $\dot{S}_{\mathrm{cg}}/(I_{\mathrm{nn}} + \varepsilon)$ is $r_s = 0.76$ ($p < 0.0001$), indicating a strong monotonic relationship. By contrast, $\mathcal{C}_{\mathrm{CCI}}$ correlates only moderately with $\dot{S}_{\mathrm{cg}}$ alone ($r_s = 0.53$) and weakly with I_{nn} alone ($r_s = 0.25$).

These results support the interpretation that the CCI captures the *balance* between coarse-grained histogram activity and structural information retention, rather than either quantity in isolation. The framework connects the structural selection programme of the companion papers to non-equilibrium thermodynamics and information theory.

1 Introduction

The Critical Coherence Index (CCI) was introduced in [4] as a dimensionless ratio

$$\mathcal{C}_{\mathrm{CCI}} = \frac{\Gamma_{\mathrm{inst}}\,\Gamma_{\mathrm{prod}}\,(1 + \kappa\, U_{\mathrm{struct}})}{\Gamma_{\mathrm{coh}} + \Gamma_{\mathrm{cons}} + \Gamma_{\mathrm{corr}} + \Gamma_{\mathrm{int}} + \varepsilon}, \tag{1}$$

designed to classify nonlinear field configurations into ordered, critical, and chaotic dynamical regimes. The index was motivated by analogy with Reynolds-type numbers in fluid dynamics: a ratio of destabilising to stabilising structural processes.

A natural question arises: does the CCI have a deeper physical grounding beyond this analogy? Specifically, can the numerator and denominator of (1) be connected to thermodynamic and information-theoretic quantities that are independently measurable?

The following hypothesis was proposed in the context of this research programme [5]:

> *The CCI approximates the ratio between the coarse-grained value-distribution turnover rate $\dot{S}_{\mathrm{cg}}$ and the structural mutual information I_{struct}: $\mathcal{C}_{\mathrm{CCI}} \approx \dot{S}_{\mathrm{cg}}/(I_{\mathrm{struct}} + \varepsilon)$.*

The physical interpretation would then be:

- **Ordered regime:** coarse-grained histogram activity small relative to structural information retention.
- **Critical regime:** comparable production and structure.
- **Chaotic regime:** coarse-grained histogram activity dominates structural organisation.

This paper tests the hypothesis numerically in a one-dimensional ϕ^4 field theory.

2 Theoretical Background

2.1 Value-Distribution Turnover Proxy

The activity diagnostic Γ_{prod} in the CCI can be interpreted as a normalised proxy for the rate at which the field value distribution rearranges over time:

$$\Gamma_{\text{prod}} \approx \frac{\dot{S}_{\text{cg}}}{S_0 + \dot{S}_{\text{cg}}}, \tag{2}$$

We stress that $\dot{S}_{\text{cg}}$ is *not* a thermodynamic entropy production rate. The ϕ^4 dynamics is a closed Hamiltonian system; by the Liouville theorem, the true phase-space entropy is conserved and thermodynamic entropy production is exactly zero. $\dot{S}_{\text{cg}}$ measures only the histogram activity — how rapidly the field value distribution rearranges — and should be understood as a proxy for dynamical activity, not dissipation. For a field $\phi(x,t)$, discretised into N_b bins, the coarse-grained entropy is:

$$S_{\text{cg}}(t) = -\sum_{k=1}^{N_b} p_k(t) \ln p_k(t), \qquad p_k(t) = \frac{|\{i : \phi_i \in \text{bin}_k\}|}{N}. \tag{3}$$

The value-distribution turnover rate is estimated as the mean absolute rate of change:

$$\dot{S}_{\text{cg}} \approx \frac{1}{T} \int_0^T \left| \frac{dS_{\text{cg}}}{dt} \right| dt. \tag{4}$$

This entropy should be interpreted as a coarse-grained information entropy proxy rather than thermodynamic entropy; it measures the rate at which the field value distribution changes over time, not a thermodynamic free energy dissipation.

2.2 Structural Information

The correlation diagnostic Γ_{corr} is interpreted as normalised structural mutual information:

$$\Gamma_{\text{corr}} \approx \frac{I_{\text{nn}}}{I_0 + I_{\text{nn}}}, \tag{5}$$

where $I_{\text{nn}} = I(\phi_i : \phi_{i+1})$ is the nearest-neighbour mutual information, measuring spatial coherence between adjacent field sites.

2.3 The Ratio Hypothesis

Under these interpretations, the dominant behaviour of the CCI is governed by:

$$\boxed{\mathcal{C}_{\text{CCI}} \approx \frac{\dot{S}_{\text{cg}}}{I_{\text{nn}} + \varepsilon}} \tag{6}$$

This ratio measures whether coarse-grained histogram activity outpaces structural information retention. A large ratio indicates that the field is losing spatial coherence faster than it can maintain structural correlations — the hallmark of chaotic dynamics.

3 Numerical Test

3.1 Setup

We test hypothesis (6) in a one-dimensional ϕ^4 scalar field theory on a periodic lattice of $N = 128$ sites:

$$\ddot{\phi}_i = \frac{\phi_{i+1} + \phi_{i-1} - 2\phi_i}{\Delta x^2} - V'(\phi_i), \qquad V(\phi) = \tfrac{1}{4}(\phi^2 - 1)^2, \tag{7}$$

integrated with a fourth-order Runge–Kutta scheme ($\Delta t = 0.02$, $T_{\max} = 24$, seed $= 42$).

Four configuration classes are studied ($n = 12$ runs each): vacuum, kink, localised, and chaotic.

3.2 Measured Quantities

For each trajectory we measure at the final state:

1. $\mathcal{C}_{\mathrm{CCI}}$ from (1) with $\kappa = 1$, $\varepsilon = 10^{-6}$.
2. $\dot{S}_{\mathrm{cg}}$ from (4) using $N_b = 20$ histogram bins and the full trajectory.
3. I_{nn} from a histogram-based estimator with 16 bins.
4. The ratio $\dot{S}_{\mathrm{cg}}/(I_{\mathrm{nn}} + \varepsilon)$.

3.3 Results

Table 1 reports the mean values per class.

Table 1: Mean ± standard deviation of CCI, coarse-grained histogram activity rate, structural mutual information, and ratio per configuration class ($n = 12$ runs, $N = 128$).

Class	$\mathcal{C}_{\mathrm{CCI}}$	$\dot{S}_{\mathrm{cg}}$	I_{nn}	$\dot{S}/I$
Vacuum	0.000 ± 0.000	0.000 ± 0.000	0.630 ± 0.095	0.000 ± 0.000
Kink	0.000 ± 0.000	0.195 ± 0.025	1.048 ± 0.084	0.186 ± 0.021
Localised	0.000 ± 0.000	0.172 ± 0.012	1.159 ± 0.071	0.149 ± 0.014
Chaotic	0.012 ± 0.003	0.181 ± 0.019	0.759 ± 0.096	0.243 ± 0.043

The correlation results are summarised in Table 2.

Table 2: Spearman and Pearson correlations between $\mathcal{C}_{\mathrm{CCI}}$ and the three candidate quantities ($n = 48$ runs total).

Quantity	Spearman r_s	Pearson r
$\dot{S}_{\mathrm{cg}}/(I_{\mathrm{nn}} + \varepsilon)$	**0.760**	**0.623**
$\dot{S}_{\mathrm{cg}}$ alone	0.530	—
I_{nn} alone	0.246	—

The ratio $\dot{S}/I$ achieves the highest correlation with the CCI ($r_s = 0.76$, $p < 0.0001$), significantly outperforming either quantity alone. This supports the hypothesis (6): the CCI captures the *balance* between coarse-grained histogram activity and structural information, not either in isolation.

4 Discussion

4.1 Physical Interpretation

The strong correlation between the CCI and the $\dot{S}/I$ ratio provides a first physical grounding for the index. The CCI may therefore be interpreted as a structural analogue of Reynolds-type instability numbers, where coarse-grained histogram activity plays the role of disorder generation and mutual information represents structural coherence. The result is consistent with the picture proposed in the context of this research programme:

- **Vacuum configurations** have zero coarse-grained histogram activity (static) and moderate structural information, yielding $\dot{S}/I \approx 0$ and $\mathcal{C}_{\mathrm{CCI}} \approx 0$.
- **Kink and localised configurations** have non-zero coarse-grained histogram activity but high structural information (extended spatial correlations), yielding a moderate ratio.
- **Chaotic configurations** have comparable coarse-grained histogram activity but lower structural information (fragmented correlations), yielding the highest ratio and CCI.

4.2 Connection to Non-equilibrium Thermodynamics

The ratio $\dot{S}_{\mathrm{cg}}/I_{\mathrm{struct}}$ connects the CCI to two established frameworks:

Prigogine's dissipative structures [6]: Prigogine's theory suggests that persistent complex patterns require a regulated level of dynamical reorganisation — neither too low (frozen) nor too high (chaotic). In the present framework, the coarse-grained histogram activity plays the role of an effective proxy for such reorganisation, while the CCI quantifies when this activity overtakes structural information retention. We note that this is an analogy: the ϕ^4 system is Hamiltonian and not a dissipative structure in the strict thermodynamic sense.

Information-theoretic approaches to complexity: structural mutual information I_{nn} measures the spatial organisation of the field. High I_{nn} with moderate $\dot{S}$ characterises ordered, information-rich configurations such as solitons.

4.3 Limitations

1. **Correlation, not identity.** $r_s = 0.76$ is strong but not perfect. The CCI and $\dot{S}/I$ are not identical; the remaining variance reflects contributions from the imbalance term U_{struct} and the coherence diagnostics Γ_{coh}, Γ_{cons}, Γ_{int}.
2. **CCI value clustering.** The CCI values for vacuum, kink, and localised configurations cluster near zero (≈ 0.000), with only the chaotic class yielding a distinctly non-zero value (0.012 ± 0.003). The reported correlation $r_s = 0.76$ is therefore driven primarily by the separation of the chaotic class from the others, rather than by a continuous trend across all classes. This warrants caution in interpreting the result as strong evidence for the hypothesis; a more stringent test would require configurations with a broader range of CCI values.
3. **Single model.** The test is performed for a 1D ϕ^4 model only. Extension to 2D, sine-Gordon, and other potentials is required to establish generality.
4. **Coarse-grained entropy estimator.** The Shannon entropy of the field value histogram is a proxy for thermodynamic coarse-grained histogram activity. More precise estimators (e.g. transfer entropy, phase-space volume measures) may strengthen the correlation.

5 Conclusion

We have tested the hypothesis that the Critical Coherence Index approximates the ratio between coarse-grained histogram activity and structural mutual information and structural mutual information in nonlinear field systems.

The Spearman correlation $r_s = 0.76$ ($p < 0.0001$) between the CCI and the ratio $\dot{S}_{\mathrm{cg}}/(I_{\mathrm{nn}}+\varepsilon)$ supports this interpretation. The result establishes that the CCI captures the balance between coarse-grained histogram activity and structural information retention — a balance that governs the transition between ordered, critical, and chaotic dynamical regimes.

This connects the structural selection programme of [1, 2, 3, 4] to non-equilibrium thermodynamics [6] and information theory [7].

Future work will extend the test to 2D field theories, driven-dissipative systems, and alternative entropy estimators, and investigate whether the correlation coefficient can be improved by refining the CCI definition.

A Reproducibility

All results are reproducible from the companion code:

https://github.com/Chris4081/structural-selection-principle

Run:

```
pip install numpy pandas matplotlib scipy
python cci_entropy_test.py
```

Key parameters: $N = 128$, $\Delta t = 0.02$, $T_{\max} = 24$, $n = 12$ runs per class, seed $= 42$, $N_b = 20$ histogram bins, $\kappa = 1$, $\varepsilon = 10^{-6}$.

References

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